

History of Information Technology

Objectives

- ◆ Define modern information technology.
- ◆ Discuss major pre-information age technologies.
- ◆ Compare and contrast early computers.
- ◆ Relate early technology to the development of modern personal computers.

Definition of Information Technology (IT)

- ◆ Use of computer hardware and software to manage information
- ◆ A combination of computing and telecommunications for the acquisition, processing, storage and dissemination of vocal, pictorial, textual and numeric information.

Electromechanical Age

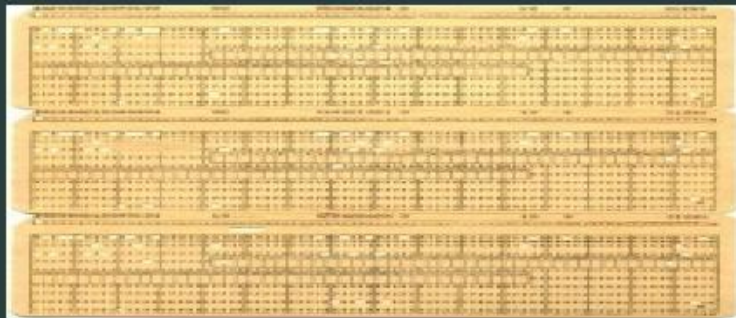
- ◆ Telegraph: invented in 1837
 - “Victorian Internet” – the telegraph was the first world communication system
- ◆ Telephone: invented by Alexander Graham Bell in 1876
- ◆ Radio: invented by Guglielmo Marconi in 1894
- ◆ These inventions could communicate information, but not store it.



Generational Technology

First generation, 1951-1958

Vacuum tubes were used as the internal computer components, punched cards and magnetic tapes for storing data, and machine language for programming.



Second generation, 1959-1964

- ◆ Transistors replaced vacuum tubes, assembly language and high level languages replaced machine language, and the removable disk pack replaced punched cards. Transistors enabled manufacturers to produce smaller computers.



- ◆ High-level programming languages were created such as Grace Hopper's COBOL, which were translated by compilers into binary format.

Third generation, 1965-1970

- ◆ Integrated circuits --complete electronic circuits on a silicon chip were developed. ICs led to the production of even smaller computers called mini computers. Software became more sophisticated and interactive processing possible



Operating systems appeared, along with the advanced programming language BASIC.

CPUs (central processing units), which contained memory, logic, and control circuits all on a single chip, personal computers (Apple II) and the graphical user interface (GUI) were developed.

GUI allows interaction with computers through images, rather having to type in commands.



Fourth generation, 1971-present

- ◆ The microprocessor or computer on a chip was developed. This made PCs, and other applications possible (calculators, banking,...)

Fifth generation, present and beyond

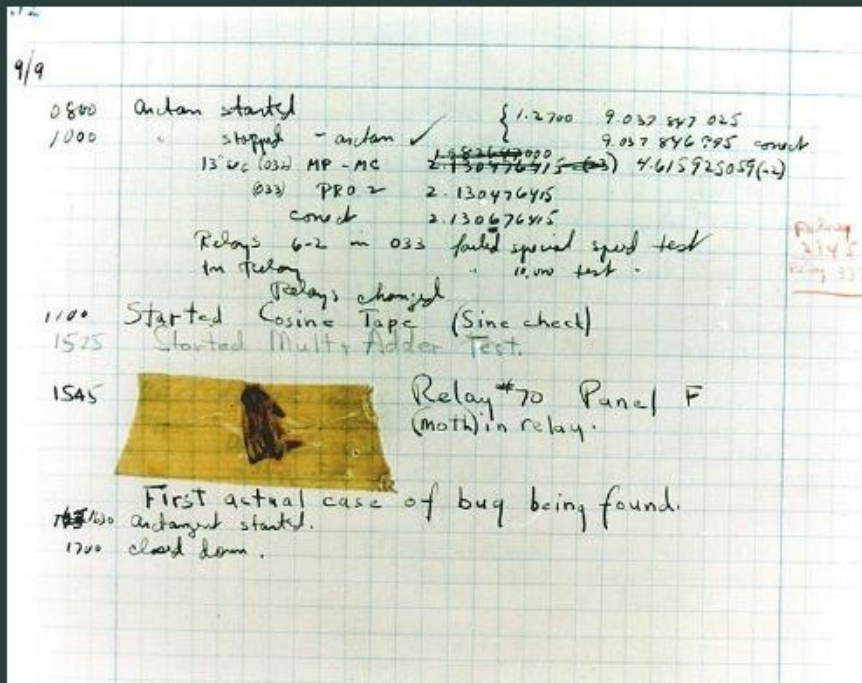
- ◆ recent and emerging technologies, i.e. voice recognition, artificial intelligence, neural systems, quantum computers...
- ◆ They used very large scale Integrated circuits.

Mark

I

- Created by Harvard student Howard Aiken in 1942

- Weighed 5 tons
- First programmable digital computer
- Used paper tape rather than punch cards
- Grace Hopper is credited with the term “debugging” when she found the first computer “bug,” a dead moth



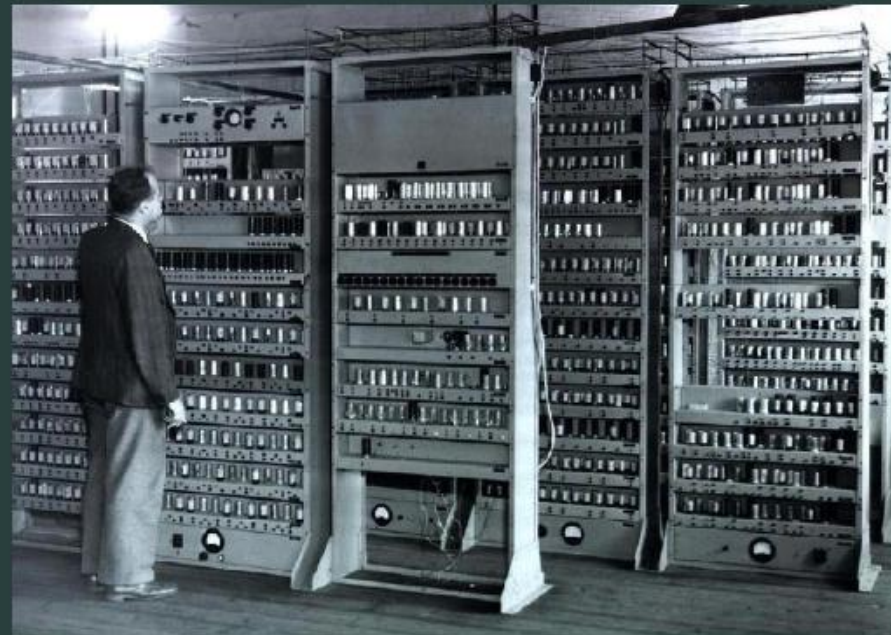
Electronic Numerical Integrator and Computer (ENIAC)

- ◆ 1st generation: used vacuum tubes
- ◆ First electronic computer
- ◆ Created by John Mauchly and J. Prosper Eckert
- ◆ Funded by U.S. Army
- ◆ Built to calculate artillery firing tables
 - Replaced female “computers”
- ◆ Could not store information
- ◆ 18,000 vacuum tubes unreliable



EDSAC

- ◆ First stored-program computer
- ◆ Invented by Mauchly and Eckert, with the help of John von Neumann
- ◆ Performed first calculation in 1949
- ◆ First graphical computer game

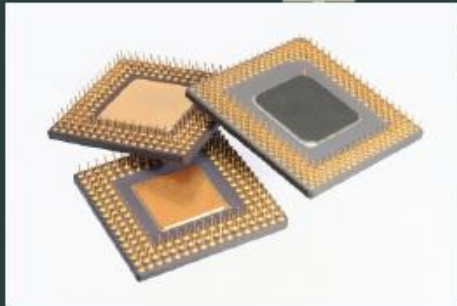


UNIVAC

- ◆ First commercially available computer
- ◆ Invented by Mauchly and Eckert
- ◆ Weighed 13 tons
- ◆ First to use magnetic tape
- ◆ Correctly predicted Eisenhower to win 1952 presidential election with a 1% population sample
- ◆ First contracts: government institutions (Census Bureau, U.S. Military branches)



Modern Technology



- 1971: Intel 4004 microprocessor developed
- ◆ 1975: Intel 8080 used in MITS Altair, first personal computer; used Microsoft's Altair BASIC software



Modern Technology, continued

- ◆ 1976: Apple I is sold as a motherboard, without a keyboard, monitor, or case
- ◆ 1981: Microsoft MS-DOS operating system
- ◆ 1984: Apple Macintosh
- ◆ 1985: Microsoft Windows



- ◆ 1993 – Multimedia desktop computers
- ◆ 1994 – Apple and IBM introduce PCs with full-motion video built in; wireless data transmission for small portable computers; first web browser invented (Mosaic)

Summar

- ◆ Information technology is the use of computer hardware and software to manage information
- ◆ Inventions such as the telegraph, telephone, and radio are used to communicate, not store, information
- ◆ Methods of data storage, retrieval, processing, and transmission change over technology generations.

Summary, continued

- ◆ The earliest computers, Mark I and ENIAC, performed calculations but could not store information.
- ◆ At first, only government institutions and corporations used computers.
- ◆ The creation of the personal computer led to the rapid development of the IT industry, which continues to grow and change today.

